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Practice Questions – Session 6 (Indexing & Slicing)

SESSION 6

1.

Create a 1D NumPy array:

```
arr = [10, 20, 30, 40, 50]
```

Print:

- First element
- Last element using negative indexing

2.

Given array:

```
arr = [5, 10, 15, 20, 25, 30]
```

Extract elements from index 1 to 4 using slicing.

3.

Given array:

```
arr = [0, 2, 4, 6, 8, 10]
```

Extract every second element using step slicing.

4.

Create a 2D array:

```
[[1, 2, 3],  
 [4, 5, 6],  
 [7, 8, 9]]
```

Print the element located at:

Row 2, Column 1 (using proper indexing).

5.

Using the same 2D array from Question 4,
Extract the first two rows.

6.

Using the same 2D array,
Extract the submatrix:

```
[[2, 3],  
 [5, 6]]
```

7.

Create array:

```
arr = [10, 20, 30, 40, 50]
```

Using fancy indexing, extract:

- 1st element
- 3rd element
- 5th element

8.

Create array:

```
arr = [5, 10, 15, 20, 25]
```

Using boolean indexing, extract elements greater than 12.

9.

Given array:

```
arr = [3, 6, 9, 12, 15]
```

Use np.where() to find indices where values are greater than 8.

10.

Create array:

```
arr = [1, 2, 3, 4, 5]
```

Reverse the array using slicing.

REVISION QUESTIONS (Previous Sessions – 2 Questions)

11. (Array Operations)

Create arrays:

```
a = [2, 4, 6]
```

```
b = [1, 2, 3]
```

Perform:

- Element-wise multiplication

- Element-wise power (a squared)

12. (Mathematical Functions)

Create array:

```
arr = [4, 9, 16, 25]
```

Compute:

- Square root
- Mean
- Maximum value

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